Fiscal Year:	FY 2025	**Task Last Updated:**	FY 10/07/2025
PI Name:	Narayanan, Anand Ph.D.		
Project Title:	Simulated Microgravity-Induced Systemic Inflammation and Its Impact on Circulatory Function and Structure		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology (2) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Project Type:	Ground	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	03/01/2019	**End Date:**	05/25/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	10	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 5/25/2025 per F. Hernandez/ARC (Ed., 8/29/24) NOTE: End date changed to 7/31/2024 per NSSC information (Ed., 11/13/23) NOTE: End date changed to 7/31/2023 per NSSC information (Ed., 1/4/23) NOTE: End date changed to 5/31/2022 per F. Hernandez/ARC and NSSC information (Ed., 6/16/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Delp, Michael Ph.D. (MENTOR: Florida State University)		
Grant/Contract No.:	80NSSC19K0426		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Long-term exposure to the space environment poses potential harm to astronaut health, including elevated morbidity and mortality risk for cardiovascular disease and heightened immune dysregulation. The cardiovascular system has adapted to gravity on Earth. The removal of hydrostatic pressure gradients during exposure to weightlessness has been thought to be the primary initiator of systemic circulatory adaptations during spaceflight. The circulation transports blood and lymph, which contain immune factors. Imbalances of immune factors can lead to increased susceptibility to infections and/or inflammatory pathologies, as well as affect the cardiovascular system. However, there is a paucity of data investigating the interplay between the cardiovascular and immune systems. This fellowship project is tied with NASA project grant 80NSSSC19K1322 (Principal Investigator Michael Delp), which has been delayed with the primary experimental effort just having completed in December 2020. Given the original project delay, I have been investigating the effects of spaceflight on the cerebral and heart circulation from Rodent-Research 9 mission (NNX16AC28G; PI Michael Delp). I am also continuing this line of research, i.e., investigating multiple organ bed vasculature adaptations to spaceflight, through a tissue sharing opportunity via International Space Station (ISS) National Lab via Rodent Reference Mission-2 (RRRM2), studying the cerebral, coronary, GI (gastrointestinal), and hindlimb vascular tissues in response to spaceflight, old age, and long-term recovery. Given delays, I have adapted the original proposal to the following aims based on the unique opportunities available to study actual spaceflight samples, in contrast to the original proposal's objectives to study simulated weightlessness adaptations. The new proposed aims include: Aim 1: To determine whether spaceflight alters the cerebral and coronary vasculature. Aim 2: To determine whether spaceflight exposure leads to adaptations in the cerebral, coronary, GI, and hindlimb vasculature in response to the single and combined effects of old age +/- recovery. Aim 3: To determine the long-term single and combined effects of simulated microgravity and/or radiation on GI and hindlimb vasculature.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	In completing these studies, I have shown how the vascular state from multiple organ beds adapts to various spaceflight conditions (actual spaceflight, simulated microgravity and/or radiation, etc.). These studies will enhance our understanding of space biological adaptations of the cardiovascular system, and furthermore, increase our understanding of medical conditions astronauts develop while in space. In studying spaceflight biomedical adaptations, we also increase our knowledge of Earth-based biomedical science by investigating novel pathways activated by the unique spaceflight environment (i.e. gravity, low-dose radiation, isolation, etc.) that may provide new perspectives of how medical conditions develop and/or biology evolve on Earth. These studies provide foundational work that I will further characterize through various on-going and future projects.
Task Progress:	Long-term exposure to the space environment poses potential harm to astronaut health, including elevated morbidity and mortality risk for cardiovascular disease and heightened immune dysregulation. The cardiovascular system has adapted to gravity on Earth. The removal of hydrostatic pressure gradients during exposure to weightlessness has been thought to be the primary initiator of systemic circulatory adaptations during spaceflight. The circulation (consisting of arteries, veins, and lymphatics) plays roles in the transport of blood and lymph, which contain immune factors. Imbalances of immune factors can lead to increased susceptibility to infections and/or inflammatory pathologies, as well as potentially alter the function of the vascular and lymphatic systems, leading to a cycle of impaired function and propagated inflammation. Spaceflight has been shown to cause immune dysregulation in astronauts, as well as increasing medical risks (such as gastroenteritis) of organ beds responsible for maintaining immune tolerance, such as the gastrointestinal (GI) system. There is a paucity of data investigating the interplay between the cardiovascular and immune systems. This fellowship project is tied with NASA project grant #80NSSSC19K1322, where I have been investigating the effects of spaceflight on the cerebral and heart circulation from Rodent Research-9 mission (Grant #NNX16AC28G). These included characterization of the molecular factors that regulate vascular function, oxidative stress state, and inflammatory factor expression in the vascular tissue. These investigations have been adapted from the original proposal of characterizing the GI and hindlimb vascular tissues, with the unique opportunity to characterize spaceflight tissues in multiple organ beds in place of simulated microgravity studies. I am also continuing this line of research, i.e., investigating multiple organ bed vasculature adaptations to spaceflight, studying the single and combined effects of deep space radiation and/or microgravity exposure on the cerebral and heart circulation, along with a tissue sharing opportunity via the International Space Station National Laboratory (ISS National Lab) via Rodent Research-2 mission, studying the cerebral, coronary, GI, and hindlimb vascular tissues in response to spaceflight, old age, and long-term recovery. These included studies of the following aims: Aim 1: To determine whether spaceflight alters the cerebral and coronary vasculature leading to molecular adaptations in vascular machinery, oxidative stress state, and inflammatory phenotype. Aim 2: To determine the long-term single and combined effects of simulated microgravity and/or radiation on the cerebral and coronary vasculature. Aim 3: To determine whether spaceflight exposure leads to adaptations in the cerebral, coronary, GI, and hindlimb vasculature in response to the single and combined effects of microgravity +/- radiation. OCTOBER 2025 UPDATE (Ed., 10/10/25) Per the Principal Investigator: • Aim 1 has been completed, manuscripts in preparation. • Aim 2 has been completed, manuscripts in preparation. • Aim 3 is ongoing, with 50% of experimental work completed, anticipating completion and submission of manuscripts, annual report, etc., by Spring 2026.
Bibliography Type:	Description: (Last Updated: 05/22/2024)